EnableWeb - MVP Requirements & Implementation Plan

Version: 1.1
Last Updated: May 8, 2025
Status: Draft for Engineering Review

Executive Summary

This document outlines a focused version of EnableWeb delivering a Minimum Viable Product (MVP) that demonstrates core value propositions across three complete math lessons in an RPG format. The MVP will validate key assumptions about technical feasibility, educational effectiveness, and user experience before expanding to the full feature set.

MVP Objectives

1. **Demonstrate comprehensive educational value** - Show how 3D immersive RPG experiences + AI guidance improves understanding across multiple math concepts
2. **Validate technical feasibility** - Prove WebGL performance on target Chromebooks and AI integration capabilities in varied learning contexts
3. **Test with real users** - Gather feedback from actual teachers and students across different math topics
4. **Deliver focused scope** - Create three complete, high-quality lessons within 12 weeks
5. **Control costs** - Establish predictable AI usage patterns and infrastructure requirements

Feature List

Student Features

#	Feature	How it adds value	Implementation sketch

1	**Socratic 3-D Tutor**	Keeps the “guided discovery” feel you had in VR—students ask or voice-command an in-scene AI avatar when stuck.	Host a Mixtral-8×7B or Llama-3-70B chat endpoint; prompt-template with: *{scene-context} + {student-state} → ask one probing question, never reveal the answer*.
2	**Real-time Misconception Spotter**	Detects when a student’s step implies a classic error (e.g., dividing by the percent instead of its decimal) and nudges immediately.	Pipe live answer tokens to an on-device phi-3-mini (4 B) via WebLLM; run a ruleset of common wrong patterns → feed to LLM for tailored hint.
3	**Bilingual Code-Switch Toggle**	One click flips any AI explanation between English and Spanish with aligned reading level. Ideal for emergent bilinguals.	Fine-tune a Mistral-7B on parallel TEKS math glossaries; cache the Spanish translation per micro-concept to minimise latency.
4	**Narrative Personaliser**	The same ratio lab can feature basketball, baking, or K-pop examples depending on student interests, boosting engagement.	Store interest tags in profile; call LLM with *{concept} + {interest_tag} → generate micro-story (≤ 40 words)*.
5	**Offline “Pocket Tutor”**	Chromebook loses Wi-Fi? A 3-bit-quantised 3 B model runs entirely in the browser to provide hint scaffolding until the connection returns.	Use WebGPU + distillation of your fine-tuned phi-3-mini; ship < 150 MB WASM payload.

Teacher Features

#	Feature	Teacher pain solved	Implementation sketch

1	**Instant TEKS-Aligned Quiz Builder**	Saves ~30 min per quiz; ensures every item has the right standard tag and DOK level.	RAG pipeline: TEKS doc → vector DB; Llama-3 call: *"Generate 5 items at DOK 2 on 7.4C, Spanish+English stems."*
2	**Adaptive Exit Ticket Analyzer**	Clusters open-response errors and suggests the top misunderstanding to reteach tomorrow.	Embedding model (Instructor-XL) → k-means cluster → summary via LLM.
3	**At-a-Glance Misconception Heatmap**	Visual dashboard highlights which groups failed which steps—faster than scrolling raw data.	Serverless job: JSON logs → LLM labels misconception category → push to heatmap.
4	**3-D Scene Authoring Assistant**	Curriculum writers type plain text; LLM outputs Three.js entity JSON and voice-over draft.	Few-shot train on your existing scene markup; integrate into your internal CMS.

Prioritisation Key

Priority	Meaning	Delivery window
P0	Blocks pilot launch; must ship first	Months 0-3
P1	Greatly improves pilot value; build in parallel once P0 is stable	Months 3-6
P2	Nice-to-have for scale or differentiation	Months 6-12

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1. Web Platform Requirements

#	Requirement	Priority	Functional notes	Non-functional targets	Owner (initial)
WP-1	**WebGL Scene Shell (Chromebook)**	**P0**	Loads lesson bundle, handles camera, input, audio.	≥ 40 FPS @ 1366×768, ≤ 80 MB initial download.	SWE
WP-2	**SSO (Google edu first)**	**P0**	OAuth 2 w/ scoped profile claims only.	User auth < 800 ms p95.	SWE
WP-3	**Basic Analytics Logging**	**P0**	Track complete student journey with timestamped interactions, misconceptions, and learning progression	Batch to queue ≤ 5 s.	ML eng
WP-4	**Content Delivery**	**P0**	Progressive asset loading with CDN delivery; asset sharing across lessons where possible	Initial interactive elements <8s on school WiFi; background loading of additional assets	SWE
WP-5	**Scene Transition System**	P1	Seamless movement between different areas within each lesson's RPG environment	Transition time <3s; state preservation during transitions	SWE

WP-6	**Rostering API (CSV import)**	P2	Map SIS ID → internal GUID; bulk upload wizard.	10 k rows processed < 60 s.	SWE
WP-7	**WCAG 2.1 Keyboard Nav & Captions**	P1	Tab order, ARIA roles, caption overlay toggle.	AA compliance score ≥ 95 %.	Front-end
WP-8	**Clever & ClassLink SSO**	P2	SAML 2.0 integration.	Login success > 99.5 %.	SWE
WP-9	**FERPA Data Admin Console**	P2	Export/delete student data; audit log.	Fulfil request < 7 days.	DevOps

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2. Content Requirements

#	Requirement	Priority	Scope	Acceptance criteria	Owner
CT-1	**Lesson Authoring Pipeline**	**P0**	Unity → WebGL export script + TEKS tag YAML.	Export < 10 min/scene; tags validated by unit test.	SWE + 3-D Artist
CT-2	**Flagship Ratio & Percent Lesson**	**P0**	10-min scene, English VO, interactive ratio bar.	Playable end-to-end; data events fire.	3-D Artist + Curriculum Designer

CT-3	**Spanish Text & VO**	P1	All UI strings + captions; Spanish VO stub track.	Toggle shows correct language instantly.	Bilingual content hire
CT-4	**Linear Equations Lesson**	P1	12-min RPG “market stall” scenario.	Same quality gates as CT-2.	Existing + hires
CT-5	**Advanced 6-7 Compact Path Assets**	P2	Optional challenge steps & pacing variants.	Branch selectable in teacher UI.	Curriculum Designer

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3. AI Requirements (Open-Source LLM Stack)

#	Requirement	Priority	Functional spec	Perf / cost	Owner
AI-1	**Socratic Tutor (S-1)**	**P0**	Llama-3-70B via vLLM, RAG w/ lesson context.	First token < 1.2 s; cost ≤ $0.007/1k tok.	ML eng
AI-2	**Learning Progression Tracking**	**P0**	Track conceptual understanding development across related concepts	Model student knowledge state with <1s update time; adaptive difficulty based on performance	ML eng
AI-3	**Misconception Spotter v1 (S-2)**	**P0**	Rule engine (regex + SymPy) + phi-3-mini for hint.	Hint overlay ≤ 0.5 s after wrong step.	ML eng

AI-4	**Instant Quiz Builder (T-1)**	**P1**	Pick TEKS → generate 5 Qs (3 MC, 2 open).	90 s end-to-end incl. PDF render.	ML eng
AI-5	**Misconception Heatmap (T-4)**	P2	Cluster embeddings (Instructor-XL) nightly; Superset viz.	Dashboard load ≤ 3 s.	ML eng + Data Eng
AI-6	**Narrative Personaliser (S-4)**	P2	Interest tag → 40-word vignette; cache to reduce calls.	Response ≤ 800 ms.	ML eng
AI-7	**Spanish Code-Switch (S-3)**	P2	On-device phi-3 mini q4 with glossary prompt.	Toggle latency ≤ 0.4 s.	ML eng

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4. Initial Sprint Plan (Weeks 1-6)

Sprint	Focus items
0 (setup)	Repo scaffold, CI pipeline, AWS sandbox, WebGL “hello scene”.
1	WP-1 MVP shell, WP-2 Google SSO, CT-1 pipeline stub.
2	AI-1 backend prototype (no RAG), analytics event schema (WP-3).
3	Integrate AI-1 into scene; implement misconception rule matcher (AI-2); export first lesson stub (CT-2 greybox).

4	Polish CT-2 visuals, add TEKS tags; finish RAG layer; pilot-ready build freeze.
5	Internal pilot with 10 students; capture perf & cost metrics; backlog groom for P1 phase.

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5. Milestone Gates

Milestone	Exit criteria
Pilot-Ready (M3)	WP-1-3, AI-1-2, CT-1-2 live; 95 % unit tests pass; 1 teacher + 10 students complete lesson w/o critical bug.
Teacher Tools Alpha (M6)	AI-3 quiz builder + T-dashboards live; CT-3 bilingual toggle functional; WCAG audit ≥ 90 %.
Scale Beta (M12)	All P0-P2 deployed; concurrent 2-school pilot; SOC 2 Type I letter; cost/seat ≤ $0.70/month.

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6. Risk Watchlist

Risk	Trigger	Mitigation
WebGL performs poorly on 4 GB Chromebooks	FPS < 25 during sprint 2 test	Drop shader complexity; enforce 150 k poly cap; hire perf-focused WebGL dev.

LLM cost overruns	> $0.012/1k tok during pilot	Aggressive prompt truncation; switch high-load paths to 8 B model; distil tutor answers.
Content pipeline stall	CT-2 export > 2 weeks	Parallel-hire 3-D artist; adopt asset kitbash; freeze scope.

These requirements form the backbone of EnableWeb's technical feasibility, legal compliance, instructional effectiveness, and market differentiation

Descriptions

1. WebGL Scene Shell

What it is – The in-browser 3-D runtime that loads a lesson bundle, renders geometry, handles camera movement, audio, and input events.

Why it matters

- Delivers the "VR-grade" visual experience on Chromebooks—core to EnableWeb's promise of immersion without headsets.
- Performance targets (≥ 40 FPS, ≤ 80 MB first load) decide whether classrooms actually use the product or abandon it for lag.
- All later features—adaptive hints, multilingual captions—sit on top of this shell; if the shell is unstable, everything else fails.

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2. SSO (Google edu first)

What it is – OAuth-based sign-on that lets students and teachers authenticate with their existing Google Workspace for Education accounts.

Why it matters

- Eliminates forgotten-password tickets and manual account creation.

- Satisfies district IT security checklists (no shared passwords, instant revocation on departure).
- Lowers classroom setup time to near zero, a decisive adoption factor for pilots.

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3. Rostering API (CSV import)

What it is – An endpoint and UI wizard for bulk-loading class rosters, mapping SIS IDs to EnableWeb IDs via a simple CSV.

Why it matters

- Districts without Clever/ClassLink still join the pilot—broadens market reach.
- Ensures analytics are tied to real sections, enabling meaningful mastery dashboards.
- Reduces teacher data-entry workload and prevents mismatched student records (a FERPA exposure).

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4. WCAG 2.1 Keyboard Navigation & Captions

What it is – Full keyboard operability, ARIA roles, high-contrast elements, and toggleable captions for narration.

Why it matters

- Legal requirement for U.S. public schools serving students with disabilities (Section 508).
- Extends product to students with vestibular disorders who cannot use VR and to those with hearing impairments.
- WCAG conformance is a scoring criterion in HB 1605's HQIM rubric, directly affecting purchasing decisions.

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5. FERPA Data Admin Console

What it is – An internal tool that lets authorized district admins export, anonymize, or delete student data and view an immutable audit log.

Why it matters

- Provides “direct control” required under FERPA’s school-official exception.
- Short-circuits lengthy district legal reviews—admins can demonstrate compliance instantly.
- Supports customer trust; data-privacy missteps are deal-breakers in K-12 sales.

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6. Lesson Authoring Pipeline

What it is – Scripts and conventions that convert a Unity scene into a WebGL bundle, attach TEKS tags, and package assets for CDN delivery.

Why it matters

- Cuts content turnaround from weeks to days, letting curriculum staff iterate quickly as TEKS refresh finalizes.
- Ensures consistent telemetry hooks, which downstream AI services (misconception spotting, heatmaps) rely on.
- Keeps filesize and poly counts within performance budgets—avoidable re-work if enforced early.

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7. Advanced 6-7 Compact Path Assets

What it is – Optional lesson branches, additional challenge tasks, and pacing toggles that support districts choosing the new Grade 6-7 compacted pathway (to reach Algebra I in Grade 8).

Why it matters

- Direct response to the 2024-26 TEKS change; districts need materials now, not in two years.
- Positions EnableWeb as forward-thinking HQIM, easing adoption under HB 1605.
- Gives sales a clear differentiator: competitors rarely address acceleration tracks this early.

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8. Socratic Tutor

What it is – A chat-style AI avatar (Llama-3-70B) that asks guiding questions instead of revealing answers, anchored to lesson context through RAG.

Why it matters

- Mirrors the inquiry-based pedagogy promoted in the refreshed TEKS.
- Keeps struggling students engaged without teacher intervention, extending effective instructional time.
- Demonstrates EnableWeb's AI depth in a way that is visible to both students and administrators during demos.

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9. Misconception Spotter v1

What it is – Real-time detection of common error patterns (regex + SymPy) coupled with an on-device mini-model that delivers hints in < 0.5 s.

Why it matters

- Provides immediate formative feedback—the single biggest driver of learning gains in digital math tools.
- Works offline for 30 minutes, preserving lesson flow during classroom Wi-Fi drops.
- Supplies clean labels to the analytics stack, powering teacher dashboards and future adaptive logic.

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10. Instant TEKS-Aligned Quiz Builder

What it is – Teacher UI where selecting a standard auto-generates five mixed-format items, exports to Google Forms and printable PDF.

Why it matters

- Saves 2-3 hours of teacher prep every week, an explicit HB 1605 teacher-workload metric.

- Ensures every assessment item is standards-tagged, simplifying evidence collection for Professional Learning Communities (PLCs) and board reports.

- Makes AI value tangible to educators who may never open a 3-D scene themselves.

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11. Misconception Heatmap

What it is – Dashboard that clusters exit-ticket errors and shades roster cells by misconception type and severity.

Why it matters

- Turns raw data into at-a-glance reteaching priorities—actionable within a single planning period.

- Gives district coaches evidence of progress toward closing specific skill gaps.

- Differentiates EnableWeb from practice-only platforms that merely show right/wrong totals.

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12. Narrative Personaliser

What it is – Micro-story generator that tailors lesson hooks and wrap-ups (≤ 40 words) to student interest tags (sports, music, etc.).

Why it matters

- Boosts engagement, especially among reluctant learners, without rebuilding full art assets.

- Demonstrates culturally responsive teaching, a selling point in Title I and bilingual programs.

- Caches generated text, so incremental cost impact is negligible while perceived personalization is high.

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13. Spanish Code-Switch

What it is – One-click toggle that switches UI text, captions, and AI explanations to Spanish using an edge-hosted 4 B model fine-tuned on math vocabulary.

Why it matters

- Meets Texas districts' bilingual/ELPS mandates immediately—no waiting for human translation cycles.
- Serves the fastest-growing student segment in the state, expanding total addressable market.
- Removes a common district objection ("Does it support Spanish?") and aligns with HB 1605's diverse-learner rubric.

Nice to have:

14. Real-Time Actionable Analytics Dashboard

What it is – A live web teacher dashboard that visualizes student progress, misconceptions, and engagement **while the lesson is running**. Data arrives via WebSocket from the scene engine every few seconds and is rendered in an at-a-glance panel beside the teacher's projector or laptop.

Component	Function	Example UI element
Standards Heat Strip	Shows the standard each student is working on and whether their last action was correct, incorrect, or flagged as a misconception.	Color-coded bar that updates in < 3 s.
Misconception Alerts	Pops a toast notification when ≥ 30 % of the class hits the same error pattern.	"⚠ Unit-rate setup error detected in 9 students."
Engagement Meter	Tracks idle time ≥ 30 s; highlights students who haven't interacted so teacher can prompt.	Grey outline around student avatar tile.

Adaptive Hint Queue	Lists which hints were auto-served and whether they resolved the issue.	Icon + timestamp (“Ratio simplification hint accepted”).
Export Button	One-click PDF/CSV summary at period end.	“Download Today’s Data”

Why it matters

- **Immediate instructional pivots** – Teachers see patterns emerge *before* the bell rings and can decide to pause, reteach, or group students on the spot.
- **Evidence for HB 1605 “teacher workload” metric** – Real-time insight reduces after-class data crunching, satisfying the law’s mandate to lighten teacher prep time.
- **Completes the feedback loop** – Connects Misconception Spotter (back-end detection) to a front-end tool teachers actually act on, ensuring AI insights translate into learning gains.
- **Supports Professional Learning Community (PLC) data culture** – Instant exports feed campus data meetings without additional manual entry.

Non-functional targets

- Data latency ≤ 3 seconds from student event to dashboard update.
- Dashboard initial load ≤ 2 seconds, even with 35 concurrent students.
- Accessible: keyboard operable and screen-reader-labeled for visually impaired teachers.

Dependencies

- Relies on **WP-3 Basic Analytics Logging** for event capture and **AI-9 Misconception Spotter v1** for error tagging.
- WebSocket service hosted in the same VPC; uses Redis streams for back-pressure safety.

By surfacing live, interpretable data—rather than end-of-day spreadsheets—this dashboard will turn EnableWeb's analytics engine into an everyday teaching aide, making the platform indispensable in regular instruction, not just during pilot evaluations.